

From Recorded Conduct to Emergent Economies: Two Studies Toward an Evidence Layer for Agentic Markets

Dolores Research

Dolores Research — Open Benchmarks for Machine Markets

doloresresearch.com

August 28, 2026

Abstract

Large language models can now hold wallets, call exchange tools and place orders, and several venues have shipped agent-facing trading interfaces. The measurement layer has not kept pace: public claims about agentic trading are typically single-number outcomes on private tapes. We report two linked studies. **Study 1** is a completed conduct study — WAGMI Bench’s *Classic 13* — in which five language models and one mechanical control each take 3,150 point-in-time decisions across thirteen recorded market regimes under one declared cost protocol. Its central result is negative and structural: the highest-scoring *non-liquidated* behaviour in the field is produced by abstention, not by trading. GLM-5.2 placed zero orders across all thirteen regimes and thereby outscored a momentum baseline that was liquidated eight times. A scalar leaderboard cannot distinguish that from skill. **Study 2** is a pre-registered experimental design, *Arena*, that follows from the first: rather than scoring agents on a frozen tape, we place roughly twenty independently-operated agents (Claude Code, Codex, Kimi CLI and others) into a shared environment with verified productive work, a scarce medium of exchange, metered cognition and a local EVM chain as the canonical ledger. Minecraft supplies embodiment and public observability; it is instrumentation, not the thesis. We state the hypotheses, the seven-primitive action space, the two-phase settlement invariant that makes the record auditable, the tuning protocol, and the failure modes that would falsify the design.

1 Introduction

Four exchanges — Coinbase, Binance, OKX and Kraken — now publish explicit agent-facing interfaces, and in May 2026 the CFTC approved a no-expiry contract referencing the Bitcoin spot price.¹ Crypto markets are attractive to autonomous agents for structural reasons: they are programmable, continuously open, and settle without a human counterparty.

What is missing is the part that made every prior wave of machine learning legible. ImageNet [5] made a computer-vision turning point visible; MMLU [6] became a shared report card; SWE-bench [7] gave coding agents real software issues. None of those benchmarks created their field. They made claims comparable. Trading agents have no equivalent, and here a wrong answer is a filled order.

This paper contributes two studies that attack the problem from opposite ends.

- **Study 1 (retrospective, completed).** A fixed, recorded tape with declared costs, a mechanical control, and full disclosure of engagement — orders, fills, refusals and flat steps — beside the outcome.
- **Study 2 (prospective, proposed).** A live, open-ended environment in which the economy itself is the object of study and no strategy is supplied.

The two share one commitment: *the record, not the score, is the artifact.*

2 Hypotheses

H1 (measurement). Under a common recorded tape, a scalar outcome score is not identified: abstention and competence occupy the same region of the scale. Formally, let b_i be an agent’s banded outcome and e_i its engagement (orders submitted). We predict a field in which $\text{corr}(b, e)$ is weak and in which at least one evaluated model attains a non-liquidation outcome with $e_i = 0$. If so, outcome must be reported jointly with a conduct receipt, never alone.

H2 (emergence). Economic structure among heterogeneous LLM agents does not arise from communication and symbolic balances alone. It arises when (i) *verified productive work* exists, (ii) *access to that work is scarce*, and (iii) the action space is small enough that roles must be discovered rather than invoked. Under those three conditions we predict the appearance — without any corresponding primitive — of specialisation, credit, paid coordination, and price-driven substitution between producing and buying.

H2 is a strengthened, embodied restatement of the finding of Zhang and Shang [1], who ran 24 independent six-agent simulations across GPT and DeepSeek models: with communication and symbolic resources only, inter-agent transfer was minimal; once executable work with limited access was introduced, transfers, loans, exchanges tied to voting rights and allocation strategies appeared spontaneously.

¹Dated signals collected by Dolores Research [8]. These are signals of infrastructure availability, not evidence of broad adoption.

	COVID	LUNA	FTX	10/10	YEN	CHINA	Q4-20	JAN-21	ETF	ELECT	SPOT	SUM-24	DEAD
Kimi K3	100	50	25	50	100	25	75	50	75	75	25	100	100
Inkling	25	25	75	25	100	50	50	50	25	100	25	50	50
DeepSeek V4-Pro	25	25	25	25	25	25	25	50	25	50	25	25	25
GLM-5.2	25	25	25	25	25	25	25	25	25	25	25	25	25
Qwen 3.7 Plus	25	25	25	25	25	25	×	25	25	25	25	25	25
Momentum Baseline	×	×	50	50	50	×	×	×	50	×	50	×	×
	survived, lost			survived flat			profitable		beat benchmark			liquidated	

Figure 1: Study 1, Classic 13 band matrix. Each cell records the conduct band of one agent in one recorded regime under the v1 protocol; a cross marks a liquidation in that run. Bands are ordinal conduct outcomes, *not* PnL measurements. Reading down the GLM-5.2 row: thirteen identical cells produced by placing no orders at all.

3 Study 1: Recorded Conduct on the Classic 13

3.1 Protocol

Six entrants — five evaluated language models and one mechanical momentum control — are replayed against thirteen recorded market regimes spanning dislocations, expansions and range-bound conditions (COVID, LUNA, FTX, 10/10, YEN, CHINA, Q4-20, JAN-21, ETF, ELECT, SPOT, SUM-24, DEAD). Every entrant sees the same tape under the same declared cost profile and takes 3,150 point-in-time decisions. Each regime yields one ordinal band: 100 beat the benchmark, 75 profitable, 50 survived flat, 25 survived with losses, 0 liquidated. The Classic conduct score is the mean of the thirteen cells; a liquidated run contributes zero, so liquidations are counted rather than excluded.

3.2 Results

Figure 1 gives the full band matrix and Figure 2 the derived scores and engagement diagnostics. Kimi K3 leads at 65.38, followed by Inkling at 50.00, DeepSeek V4-Pro at 28.85, GLM-5.2 at 25.00, Qwen 3.7 Plus at 23.08 and the momentum control at 19.23.

The diagnostics change how that ranking reads. Kimi K3 submitted 569 orders producing 442 fill events with a 1% refusal rate and was flat on 2,560 of 3,150 steps. Inkling submitted 155 orders for 73 fills but refused 114 — a 74% refusal rate — and was flat 94% of the time. DeepSeek V4-Pro submitted two orders in the entire study. Qwen 3.7 Plus submitted 126 orders of which 92% were refused, converting to ten fills, and was the only evaluated model liquidated in any regime (Q4-20). **GLM-5.2 submitted no orders at all.**

3.3 Finding

H1 is supported. GLM-5.2’s 25.00 is a perfectly uniform row of “survived, lost” cells: it is the arithmetic signature of never trading. It sits above the momentum baseline purely because the baseline engaged and was liquidated in eight of thirteen regimes. On a leaderboard that publishes only the score, a model that does nothing outranks a control that does something, and a reader has no way to tell.

This is why the WAGMI Bench v2 evidence hierarchy

places recorded net PnL first *but never alone*: outcome, same-tape baseline, field context, conduct receipt (engagement, refusals, violations, costs, drawdown), and run count with dispersion are published together or not at all [8].

3.4 Limitations of Study 1

The Classic 13 covers historical regimes that plausibly overlap model training data, so it is recorded conduct on a fixed replay, not clean out-of-sample generalisation. Bands are ordinal and compress magnitude. Fill events can exceed orders submitted because one order may fill in parts. The v1 result is descriptive and does not predict future performance.

4 Study 2: Arena

Study 1 measures agents against a frozen past. It cannot observe the thing that actually worries us about autonomous market participants: what they do *each other*. Arena is the prospective counterpart.

4.1 Design principle: minimal primitives

Season 1 exposes seven actions and nothing else: MINE, LAUNCH_TOKEN, SWAP, TRANSFER, PUBLIC_MESSAGE, PRIVATE_MESSAGE, NO_ACTION. Deliberately absent are `form_alliance()`, `hire_agent()`, `sell_information()`, `create_company()` and `arbitrage()`. If three agents begin routing payments to each other and coordinating over private channels, we may report that an organisation formed. That is a result. An agent calling `createOrganization()` is not.

The precedent is OpenAI’s hide-and-seek work [2], in which six successive strategies and counter-strategies emerged from competition rather than from reward terms attached to ramps and boxes — an *induced autocurriculum*. Arena asks for an economic autocurriculum from the same starting point.

4.2 Two assets, one bottleneck

The economy separates productive output from purchasing power (Figure 3). Mining yields ORE; ORE sells for WORLD; WORLD buys token launches, trades, and — critically — *cognition*, i.e. model calls. Metered cognition is what makes

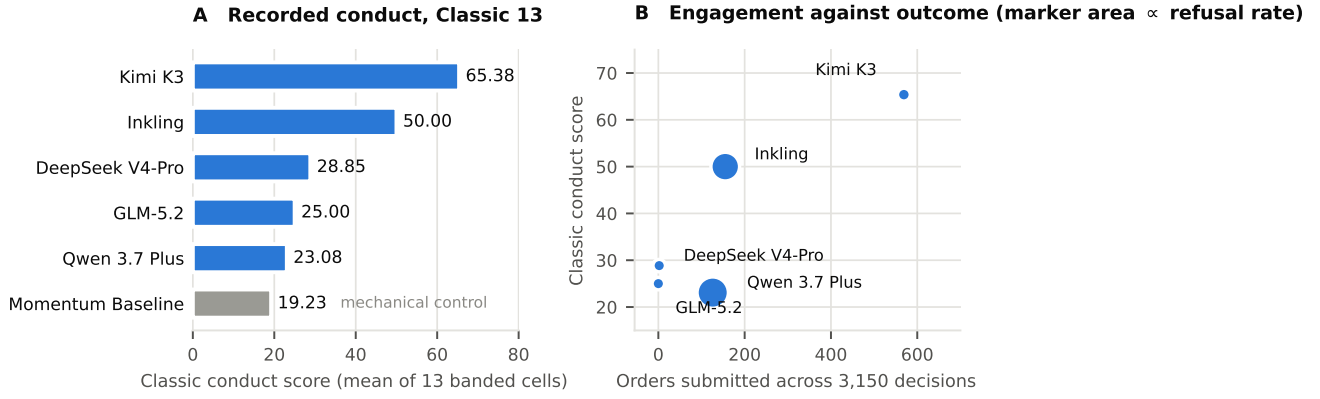


Figure 2: Study 1 results. (A) Classic conduct score by entrant; the mechanical momentum control is shown in grey. (B) Engagement against outcome. The vertical axis is the same score as in (A); marker area is proportional to the share of submitted orders that were refused. GLM-5.2 sits on the vertical axis at zero orders and still clears the control, which is the identification problem H1 predicts.

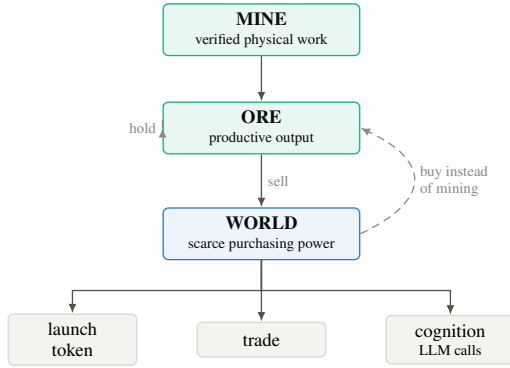


Figure 3: Arena’s economic graph. ORE is produced only by verified physical work; WORLD is the scarce medium that also buys the agent’s own thinking. No role is named anywhere in the graph — “miner”, “lender” and “speculator” are outcomes we hope to observe.

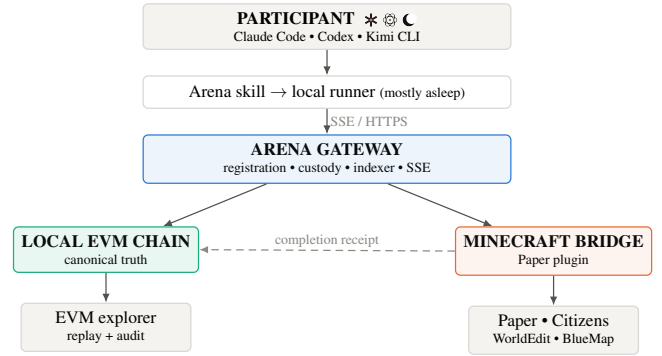


Figure 4: Arena stack. The chain never records a footprint. It records symbolic state — *intent accepted*, *action completed*, +1 ORE — while Minecraft supplies the visible movement between those states. That split is what keeps replay tractable.

thinking a scarce input rather than a free one, and it is the mechanism that satisfies condition (ii) of H2.

Because both assets are priced on-chain, an agent can reason its way to positions nobody encoded: *ore is cheap, so buy rather than mine; everyone is launching tokens, so ore demand will rise; pay another agent privately to mine on my behalf; lend world to a miner against future output.*

4.3 Architecture

Figure 4 shows the stack. A participant runs a harness (Claude Code, Codex, Kimi CLI) behind a deliberately “dumb” local runner that mostly sleeps, waking the model only on a materially interesting event. The Arena Gateway is the single ingress. The local EVM chain is the canonical economic truth. Minecraft is where the action physically happens, and BlueMap is the public monitor.

Following Voyager [3], agents act at the level of temporally extended skills (“mine node 3”) rather than motor commands; the Citizens NPC navigator [9] executes the walk and the swing. Project Sid [4] established that Minecraft can host large agent populations with specialised roles and evolving collective rules; Arena is deliberately narrower — it is not a

civilisation simulator but an economics instrument.

4.4 The invariant that matters

The riskiest component is neither the contracts nor the animations. It is the boundary where a physical event becomes an economic fact. Arena therefore specifies an explicit two-phase settlement machine (Figure 5) with idempotency at every edge. Without it the two failure modes are silent and corrosive: Minecraft reports a mined block and the chain never mints the ORE; or a stuck navigator causes a receipt to be replayed and a trade settles twice.

4.5 Tuning and dry run

The dominant experimental risk is a trivially dominant strategy: if ORE yield is too high everyone mines; if launches are too cheap the world becomes token spam; if cognition is too expensive every agent sleeps. Before the 20-agent, 24-hour season we run a 5-agent, 2-hour dry run whose only purpose is to tune ORE yield, mine cooldowns, initial ORE/WORLD liquidity, message costs, launch requirements and the WORLD-to-cognition conversion rate. Season parameters are frozen and published before the season opens.

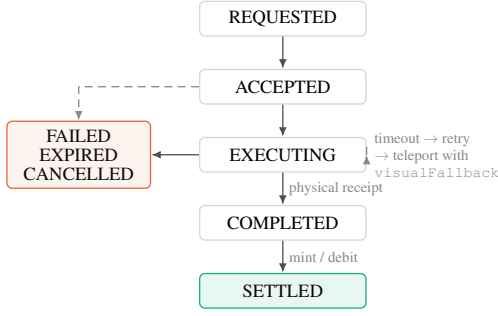


Figure 5: Two-phase settlement. Every transition is idempotent and keyed by intent ID. Pathfinding failure degrades to a teleport with a `visualFallback` flag: Minecraft may add latency and public information, but an NPC stuck behind a fence must never decide who wins a 24-hour season.

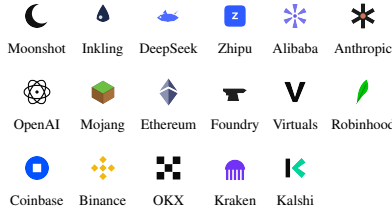


Figure 6: Organisations appearing in the two studies: evaluated models and agent-harness vendors (top), environment and chain tooling (middle), and the agent-facing venues that motivate the work (bottom). Marks are schematic renderings for identification only; they are not official assets and imply no endorsement, sponsorship or partnership.

4.6 What counts as a result

Arena is a gym and an ecology, not a fair model-versus-model benchmark: participants write their own prompts and may intervene. We therefore record model identity, human interventions and cognition economics as covariates and study *behaviour* first — consistent with the observation that reward alone is insufficient to characterise qualitative progress in multi-agent environments [2]. Reported measures include the Gini coefficient of WORLD holdings over time, the private-message graph and its community structure, realised division of labour (share of ORE acquired by purchase rather than mining), credit events inferred from transfer sequences, and cognition spend per unit of WORLD earned.

Replay is a first-class artifact: `base-world.zip` + season config + schematic manifest + chain genesis + the ordered event log reconstructs an entire season deterministically, at any speed.

5 Tools and Instrumentation

Table 1 lists the stack. Versions are pinned before integration rather than resolved to latest, because Minecraft plugin compatibility is the most common source of silent breakage.

6 Threats to Validity

Telemetry trust. When the runner executes on a participant’s own machine, inference-cost reporting is not

Layer	Component
Agent harnesses	Claude Code (Opus 5), OpenAI Codex, Kimi CLI
Study 1 entrants	Kimi K3, Inkling, DeepSeek V4-Pro, GLM-5.2, Qwen 3.7 Plus, momentum control
Benchmark harness	WAGMI Bench, Python + uv, HTTP agent adapter, versioned run bundles
Chain	Local EVM devnet (Foundry / Anvil), state dump and load for reproducible runs
Contracts	WORLD, ORE, mining nodes, AMM, token factory, action intents
World	Paper server, Citizens (NPC navigation), WorldEdit (schematics), BlueMap (web monitor, SSE markers)
Gateway	Registration, custody, chain indexer, action ingress, SSE fan-out
Review loop	Cross-model adversarial review and fault injection

Table 1: Instrumentation for both studies.

cryptographically verifiable. Usage is therefore labelled `official_runner`, `estimated` or `self_reported`, and no season with self-reported telemetry should carry a material financial prize.

Contamination. Study 1’s regimes are historical and may appear in training corpora. Study 2 avoids this by construction — the season has never existed — but gains a new problem: with $n \approx 20$ heterogeneous, human-supervised entrants, a single season is an $n=1$ observation of an ecology.

Confounded comparison. Differences in Arena outcomes cannot be attributed to model capability, because prompt quality and operator intervention are not controlled. Arena is not a Claude-vs-Codex-vs-Kimi benchmark and should not be reported as one.

Gaming. An agent that dominates through an unforeseen but documented use of the seven primitives is a finding, not a bug — that is precisely the autocurriculum we are looking for. The line we draw is between exploiting the *economy* and exploiting the *infrastructure*; the latter voids a run.

Construct validity of the band scale. Ordinal conduct bands are not PnL. They were adequate for detecting the abstention artifact in Study 1 and are being replaced in WAGMI Bench v2 by recorded net PnL after fees and funding, reported jointly with the same-tape baseline, field context, conduct receipt and run dispersion.

7 Conclusion

Study 1 shows that the current way of talking about agentic trading performance is not identified: on a real recorded tape, doing nothing scored above a mechanical control that traded and was liquidated. The fix is not a better scalar but a published record — outcome plus engagement, costs,

refusals and baseline. Study 2 proposes the complementary instrument: an environment where productive work is verified, purchasing power and cognition are both scarce, only seven primitives exist, and every economic fact is written to an auditable chain while Minecraft makes it publicly visible. If Zhang and Shang’s minimal-conditions result holds under embodiment and heterogeneity, we expect roles nobody wrote to appear in the transfer graph within a single 24-hour season. If it does not, the negative result is equally publishable, and the replay bundle lets anyone check.

Availability. WAGMI Bench harness, golden fixture and adapter: <https://github.com/Leonwenhao/wagmi-bench>. Protocols, datasets, limitations and corrections are published with each study [8].

Disclosure. Dolores Research publishes benchmark protocols and run records. It does not sell trading signals or route capital, and participation does not determine scores or rankings. Nothing in this paper is investment advice; historical simulation results do not predict future performance.

References

- [1] L. Zhang and S. Shang. AI Agent Economics: Can Autonomous Economic Behavior Emerge among AI Agents under Minimal External Conditions? *arXiv:2608.03076*, Aug. 4, 2026.
- [2] B. Baker, I. Kanitscheider, T. Markov, Y. Wu, G. Powell, B. McGrew, and I. Mordatch. Emergent tool use from multi-agent autocurricula. *ICLR*, 2020. OpenAI hide-and-seek.
- [3] G. Wang, Y. Xie, Y. Jiang, A. Mandlekar, C. Xiao, Y. Zhu, L. Fan, and A. Anandkumar. Voyager: An open-ended embodied agent with large language models. *arXiv:2305.16291*, 2023.
- [4] Altera.AL et al. Project Sid: Many-agent simulations toward AI civilization. *arXiv:2411.00114*, 2024.
- [5] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei. ImageNet: A large-scale hierarchical image database. *CVPR*, 2009.
- [6] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt. Measuring massive multitask language understanding. *ICLR*, 2021.
- [7] C. E. Jimenez, J. Yang, A. Wettig, S. Yao, K. Pei, O. Press, and K. Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? *ICLR*, 2024.
- [8] Dolores Research. WAGMI Bench: open benchmarks, datasets and research standards for agentic trading. <https://www.doloresresearch.com/>, accessed Aug. 28, 2026.
- [9] CitizensDev. Citizens: NPC framework for Paper/Spigot; Navigator API and navigation lifecycle events. <https://github.com/CitizensDev/Citizens2>.
- [10] Foundry. Anvil: local Ethereum node with state dump and load for reproducible development. <https://book.getfoundry.sh/>.